

RESEARCH ARTICLE OPEN ACCESS

Ultralow-Power Real-Time On-Chip Thermal Prediction via Finite Element Method–Machine Learning Codesign and Field-Programmable Gate Array Deployment

Jun Ho Lee | Jae Gyu Kim | Sung Jin Kim | Jae Yong Song | Byoung Don Kong 

Pohang University of Science and Technology (POSTECH), Pohang, Korea

Correspondence: Byoung Don Kong (bdkong@postech.ac.kr)

Received: 10 September 2025 | **Revised:** 20 December 2025 | **Accepted:** 15 January 2026

Keywords: advanced packaging | field-programmable gate array acceleration | finite element method | heat-map prediction | machine learning | Pareto optimization | thermal management

ABSTRACT

Thermal management has become a critical challenge in advanced semiconductor packaging, where high power density and complex three-dimensional integration hinder direct temperature measurement and increase the risk of performance degradation. This study presents a scalable and power-efficient framework for real-time, die-level temperature prediction by integrating finite element simulations, machine learning (ML), and hardware acceleration on a reconfigurable processor. A compact thermal model was first calibrated using limited infrared thermography data through hybrid ML–Pareto optimization, ensuring balanced accuracy across multiple power conditions. The optimized simulations generated high-fidelity synthetic datasets, which—combined with empirical measurements—were used to train a lightweight single-layer perceptron capable of predicting heat maps from surface-temperature inputs. Implemented on a Xilinx Zynq platform, the model achieved inference latencies comparable to a desktop GPU while consuming only 1.71 W—delivering more than 21 times greater energy efficiency than the GPU and over 2600 times faster computation than full finite element simulations. The proposed approach enables accurate, real-time thermal monitoring in resource-constrained environments, with strong potential for deployment in edge artificial intelligence, mobile systems, and Internet of Things devices. This framework also opens pathways for integrating predictive thermal control into next-generation intelligent electronic systems.

1 | Introduction

With the continued scaling of semiconductor processes, advanced packaging technologies such as through-silicon vias (TSVs) and microbumps are increasingly adopted to enable high-performance, high-density integration [1, 2]. These approaches support multilayered architectures and heterogeneous integration within a single package, significantly boosting system capability [3]. However, high-density stacking also raises power density, exacerbates thermal management challenges, and increases the risk of manufacturing defects [4–8]. In three-dimensional architectures, TSVs provide vertical interconnects but restrict heat dissipation paths, elevating internal die temperatures and potentially

degrading performance or reducing device lifetime [9]. As process nodes shrink, variability and defects—such as misaligned microbumps or irregular TSV fabrication—become more common, often inducing localized heating and, in severe cases, thermal failure during operation [10, 11].

Postfabrication thermal management is therefore essential for maintaining reliability in advanced packages. Once the chip is encapsulated with a heat spreader or lid, however, direct measurement of internal die temperatures is no longer feasible [12, 13]. Embedded temperature sensors provide only sparse [14], point-wise measurements (tens per chiplet vs. thousands of potential hotspots), leaving large areas unmonitored. This makes it

This is an open access article under the terms of the [Creative Commons Attribution](https://creativecommons.org/licenses/by/4.0/) License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

© 2026 The Author(s). *Advanced Intelligent Systems* published by Wiley-VCH GmbH.

extremely difficult to capture localized gradients or predict hot-spot evolution using conventional sensing strategies alone [15]. A new approach is required—one that enables high-resolution, real-time monitoring and control of internal die temperatures during active operation, even under lid-enclosed conditions.

Recent research has shown the potential of machine learning (ML) to predict chip temperatures far more efficiently than conventional finite element method (FEM) or finite-difference simulations. Numerous studies have demonstrated that, given accurate power and boundary condition data, ML-based thermal solvers can reconstruct steady-state or transient heat maps with orders-of-magnitude speedup over classical methods [16, 17]. However, most of these works have remained at the theoretical or software-simulation level, without direct integration into practical, on-device monitoring systems.

In this work, we move beyond simulation and present the first end-to-end implementation of an ML-based thermal-prediction framework for practical advanced packaging applications. Specifically, we target real-time prediction of internal chiplet temperatures in lid-sealed environments. Our approach integrates FEM-based compact thermal modeling (CTM) with Pareto-optimized ML training and deploys the resulting neural network directly onto a field-programmable gate array (FPGA) for in situ inference. During calibration, lid-off infrared (IR) thermography provides ground-truth internal temperature distributions under multiple operating voltages, which are then used to refine FEM simulations and generate large-scale, physically consistent datasets. Once trained, the framework requires only lid-surface temperatures as input, ensuring compatibility with realistic packaged-device conditions while retaining the fidelity of calibration data.

Beyond runtime monitoring, this strategy also offers potential for chiplet product quality control. By lightly operating chips

without a lid during testing, one can compare expected versus observed heat distributions from IR measurements, thereby diagnosing poor thermal contacts, voids, or defective microbumps before full packaging. Such capability is particularly valuable in chiplet systems, where the reliability of thousands of interconnects and interfaces must be ensured for yield.

In addition to prediction accuracy and inference speed, our approach emphasizes low-power operation—a critical requirement in mobile and edge artificial intelligence (AI) platforms where thermal and power budgets are highly constrained [18, 19]. While traditional FEM simulations provide high-fidelity thermal data, they are time-consuming and computationally expensive [20, 21]. Likewise, GPU-based inference can deliver high throughput but consumes tens of watts during continuous operation [22, 23]. In contrast, our FPGA-based implementation achieves real-time inference with under 2 W power consumption, offering more than a 21-fold improvement in energy efficiency over commercial GPUs. This hardware-ready approach transforms a promising research concept into a deployable thermal management tool for advanced packaging, enabling both operational monitoring and manufacturing diagnostics in environments where reliability, latency, and energy efficiency are critical.

2 | Methodology

The proposed framework was implemented through a four-stage workflow (Figure 1). First, IR thermography was performed under four power conditions (P1–P4) to obtain reference temperature data. These measurements were used to calibrate an FEM model via Pareto-based multiobjective optimization, yielding a CTM with physically consistent parameters.

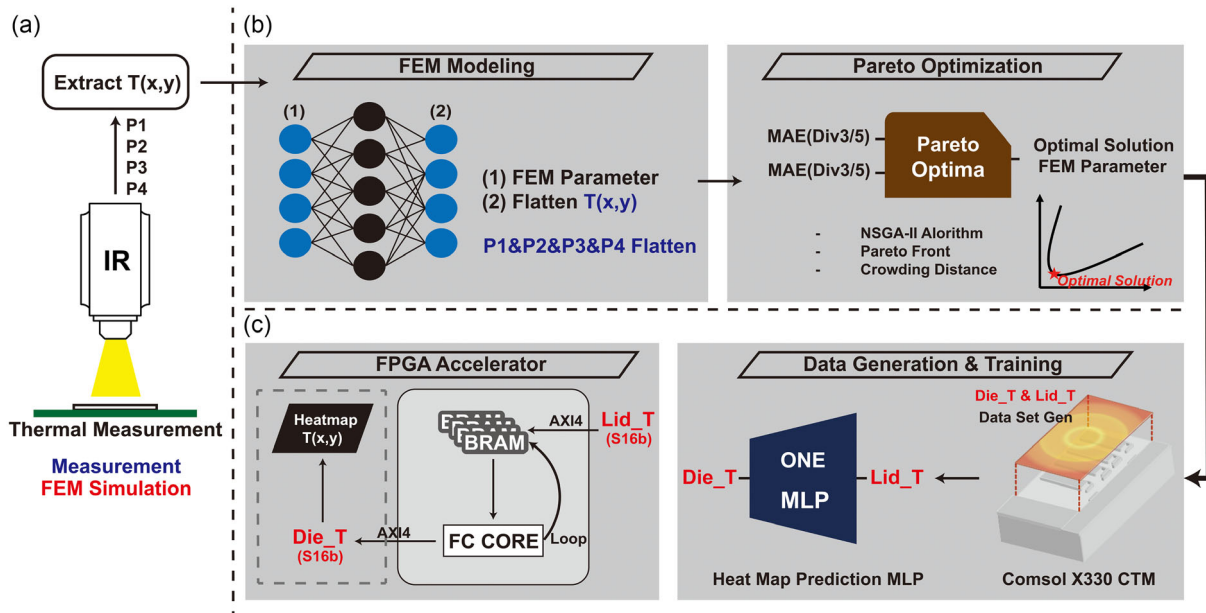


FIGURE 1 | Four-stage workflow of the proposed FPGA-accelerated on-chip thermal-prediction framework. Stage 1: Infrared (IR) thermography under four power conditions (P1–P4) provides lid and die-temperature maps for model calibration and validation. Stage 2: Pareto-based multiobjective optimization calibrates an FEM CTM with physically consistent parameters. Stage 3: The calibrated FEM model generates high-resolution synthetic datasets for supervised training of an ML predictor that maps lid-surface temperatures to die-level heat maps. Stage 4: The trained predictor is implemented as a parallel fully connected (FC) core on an FPGA, achieving real-time inference within a ~2 W power envelope.

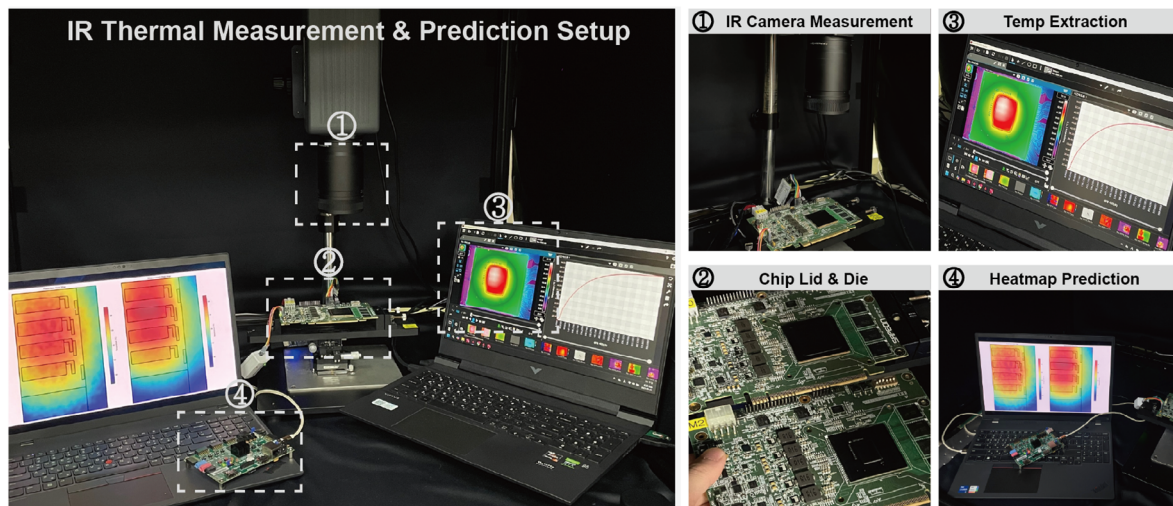


FIGURE 2 | IR thermography setup and lid-on/lid-off measurement procedure for generating calibration data. (1) Test platform comprising a Sapeon X330 chip mounted on a custom evaluation board, with an overhead IR camera capturing spatial temperature distributions. (2) Lid-on configuration used to record surface (lid) temperatures under operating power conditions. (3) Lid-off configuration exposing the die for direct measurement of internal temperature maps, serving as ground-truth references for FEM calibration and ML training. (4) Data-processing pipeline converting raw IR images into aligned temperature fields $T(x, y)$ for FEM fitting and heat-map prediction.

Next, the calibrated FEM model was used to train an ML predictor capable of converting lid-surface-temperature maps into die-level heat maps. The trained network was then mapped onto a parallel fully connected (FC) core and deployed on an FPGA, enabling real-time inference within a ~ 2 W power envelope.

The experimental setup (Figure 2) employed a Sapeon X330 test chip mounted on a custom evaluation board. An overhead IR camera recorded lid-temperature distributions during normal operation, followed by lid-off measurements to directly capture die temperatures for calibration and validation. These two-dimensional temperature fields, $T(x, y)$, were processed on a host workstation, supplied to the FEM for parameter fitting, and used as supervised learning targets for the prediction network.

Although IR thermography provided accurate references, its throughput was limited—each cycle required nearly an hour for stabilization, calibration, and repetition, and the lid-on/lid-off procedure doubled the effort. As a result, the number of measured samples was insufficient for robust ML. To overcome this constraint, the calibrated FEM model was employed to generate a synthetic dataset with high spatial resolution and physical consistency, enabling comprehensive coverage of operating conditions and balanced training across all power levels.

2.1 | Finite Element Modeling for Data Augmentation

Developing an ML model capable of accurately predicting on-chip temperature distributions requires a comprehensive thermal dataset across diverse power conditions. However, direct acquisition of such data via IR thermography is highly time- and cost-intensive, as each condition requires nearly 1 h for thermal stabilization, calibration, and repeated measurement. Furthermore, our setup requires both lid-on and lid-off configurations, effectively doubling the acquisition effort. Extended measurement durations may also introduce environmental noise, reducing overall data reliability.

To overcome these limitations, FEM was adopted to produce synthetic yet physically grounded thermal maps, addressing the limited volume of measured data. FEM is particularly suited to advanced packaging environments containing heterogeneous elements such as TSVs and microbumps, as it can capture complex three-dimensional heat-transfer pathways with high spatial fidelity [24]. The calibrated FEM model not only reproduced the experimentally measured distributions but also expanded the dataset to conditions that could not be practically obtained through IR acquisition alone, forming the basis for large-scale model training.

Temperature distributions measured under four representative power conditions (P1–P4) were used to calibrate a CTM within COMSOL Multiphysics. The Sapeon X330 is a bare-die prototype device without packaging or a heat spreader, as direct IR thermal imaging of the die surface is required. This configuration significantly constrains the allowable operating power range due to increased sensitivity to overheating.

Through experimental evaluation, we identified a minimum operating power below which temperature variations are indistinguishable from ambient noise and a maximum power beyond which sustained operation risks thermal failure. Within this bounded safe operating region, stable benchmark workloads could only be executed at a limited number of operating points. The four power conditions—5.23 W/1.39 W (P1), 3.56 W/1.26 W (P2), 2.72 W/1.14 W (P3), and 2.34 W/1.10 W (P4), corresponding to NPU and VDD powers, respectively—were therefore selected as reliable reference conditions for thermal calibration. These power levels are not evenly spaced, as they are defined by benchmark feasibility and hardware safety constraints rather than by uniform power division.

To supplement the measured power–temperature dataset within this safe operating window, additional intermediate power conditions were generated using COMSOL simulations by sweeping the NPU and VDD powers in 0.25 W and 0.03 W increments, respectively. These interpolated operating points provided a denser set of temperature profiles for model training.

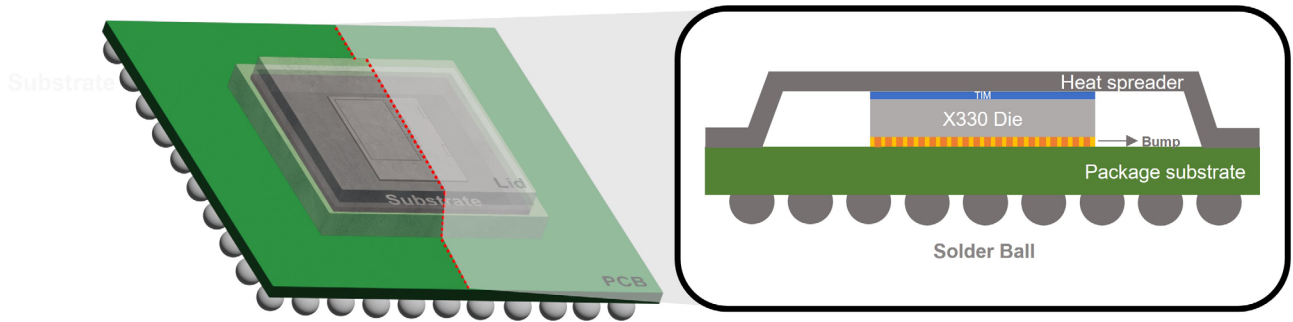


FIGURE 3 | Schematic of the test package and cross-section showing the heat spreader, X330 die, bumps, and package substrate on a PCB.

CTM calibration was performed using multiobjective optimization, during which key thermal parameters—including heat-transfer coefficient, thermal conductivity, heat capacity, and surface emissivity—were adjusted to minimize discrepancies between simulated and measured temperature distributions. Ambient conditions and heat-source localization were configured to match the experimental measurement setup.

The FEM geometry was based on the physical dimensions and material properties of the Sapeon X330 package, including the heat spreader, silicon die, microbumps, and substrate (Figure 3). Two major functional regions, the NPU and VDD blocks, were modeled as distinct heat sources to capture the asymmetric power dissipation observed experimentally. Using the “Heat Transfer in Solids” module, volumetric power inputs were applied as:

$$Q_0 = \frac{P_0}{V} \quad (1)$$

where P_0 is the applied block power (W) and V is block volume (m^3) [25]. Heat transport was governed by:

$$\rho C_p u \cdot \nabla T + \nabla \cdot q = Q + Q_{\text{ted}} \quad (2)$$

with ρ the density (kg/m^3), C_p the specific heat ($\text{J}/\text{kg}\cdot\text{K}$), u the velocity vector (for any defect-driven flow), k the thermal conductivity ($\text{W}/\text{m}\cdot\text{K}$), Q the internal heat generation, and Q_{ted} additional sources such as thermoelectric effects [25]. Boundary heat loss via convection/radiation was modeled as:

$$q_0 = h \cdot (T_{\text{ext}} - T) \quad (3)$$

where h is the heat-transfer coefficient ($\text{W}/\text{m}^2\cdot\text{K}$), T_{ext} is ambient temperature, and T is the chip surface temperature [26].

The calibrated FEM model was validated against measured IR data, showing close agreement in both absolute temperature and spatial distribution. Once validated, it was used to generate high-resolution surface maps (61×45 grid) spanning the full voltage–power design space. These synthetic datasets were combined with measured data to train the ML-based heat-map predictor, ensuring balanced coverage and improved generalization.

2.2 | Data Fitting to Measured Values

To bridge the gap between IR measurements and high-fidelity FEM simulations, we adopted a hybrid calibration strategy that combined ML with multiobjective Pareto optimization. Starting

from baseline material properties in COMSOL, a parameter sweep generated ~ 1000 synthetic temperature maps, which were used to train an initial ML surrogate. This surrogate enabled inverse inference to estimate four key thermal parameters—heat-transfer coefficient, silicon thermal conductivity, heat capacity, and surface emissivity—as illustrated in the neural network architecture shown in Figure 4.

However, as summarized in Table 1, the ML-only fitting exhibited an imbalance in prediction accuracy across the four representative power conditions (P1–P4), each measured as 61×45 temperature maps (10 980 points). Specifically, the model tended to overfit to either the (P1 & P3) group or the (P2 & P4) group, leading to degraded performance in the opposite cases. This imbalance highlights a key limitation: parameter tuning based on a single objective cannot ensure uniform accuracy across all operating conditions. To address this issue, a Pareto-based multiobjective optimization was introduced. The prediction errors were grouped into two categories to better capture and balance the model’s tendency to underperform under different thermal scenarios.

To quantify this imbalance, mean absolute error (MAE) was aggregated as:

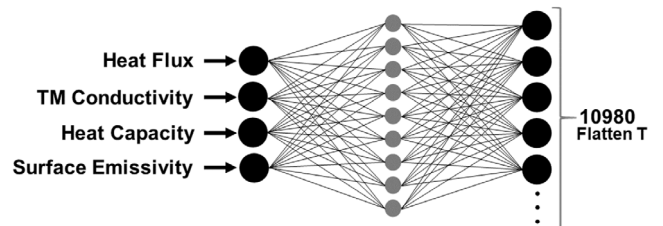


FIGURE 4 | Surrogate neural network architecture for predicting the 2D chip temperature distribution from thermal parameters.

TABLE 1 | Comparison of measured and ML-predicted temperature ranges under four power conditions (P1–P4), with direction of deviation indicated.

	Measurement Temp, °C	ML Temp, °C	Deviation
P1	64.1–60.0	64.2–60.0	↓
P2	57.1–54.0	56.7–54.0	↑
P3	52.3–50.2	52.5–50.5	↓
P4	51.5–49.5	51.0–49.2	↑

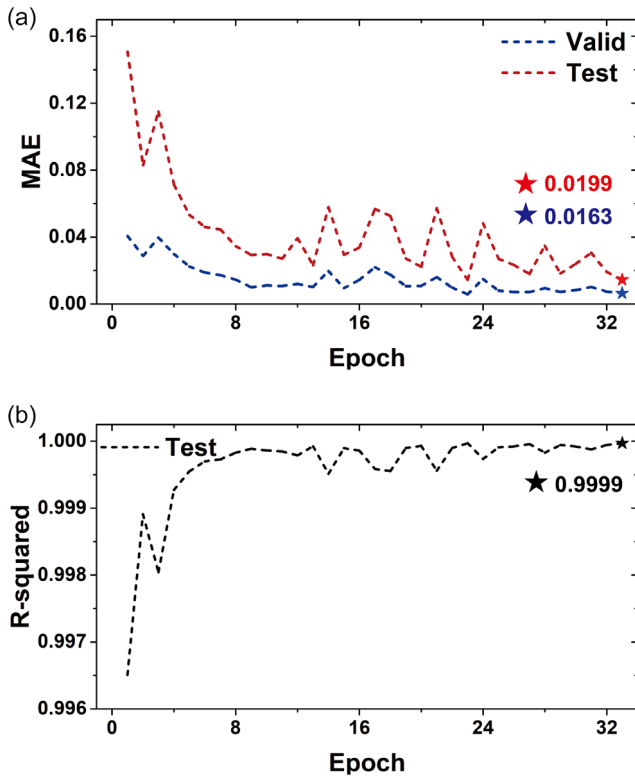


FIGURE 5 | (a) MAE trend for validation and test datasets. (b) Test R-squared trend showing stable convergence.

$$\text{MAE SUM1} = \text{MAE}[P1] + \text{MAE}[P3] \quad (4)$$

$$\text{MAE SUM2} = \text{MAE}[P2] + \text{MAE}[P4] \quad (5)$$

ML-only tuning often minimized one sum at the expense of the other. To enforce balance, the optimization targeted:

$$\text{MAE SUM1} \approx \frac{1}{\text{MAE SUM2}} \quad (6)$$

which prevents improvements in one group from excessively degrading the other.

This formulation ensured that improvements in one group did not excessively deteriorate the performance of the other, thereby promoting balanced calibration across the entire power range. To achieve such balanced trade-offs, dense sampling of the parameter space was required. In total, a fine-grained sweep over four key parameters—heat-transfer coefficient, silicon thermal conductivity, heat capacity, and surface emissivity—yielded 68 452 combinations. However, since each FEM simulation took ≈ 72 s, a brute-force evaluation of all candidates would require over 57 days, making it infeasible for iterative design. To overcome this limitation, we adopted a “sampling + ML surrogate” strategy. A subset of 4000 parameter combinations were simulated, and the resulting thermal profiles were used to train a DNN surrogate in about 880 s, enabling ≈ 14 ms predictions—about 17 times faster than direct FEM.

After 34 epochs, the surrogate reached a test coefficient of determination of $R^2 = 0.9999$, with a validation MAE of 0.01999 and a test MAE of 0.0163, achieved within an average epoch time of 40 s. These results, consistent with Figure 5, confirmed that the model converged stably without any indication of overfitting.

TABLE 2 | Computational time comparison for generating 68 452 thermal samples using FEM-based COMSOL simulations versus ML-based inference.

Data Num	COMSOL [MATLAB]		ML	
68 452	One data simulation time	72 s	Training time	880 s
			One data prediction time	0.014 s
	All data generation time	4 928 544 s	All data generation time	1838 s

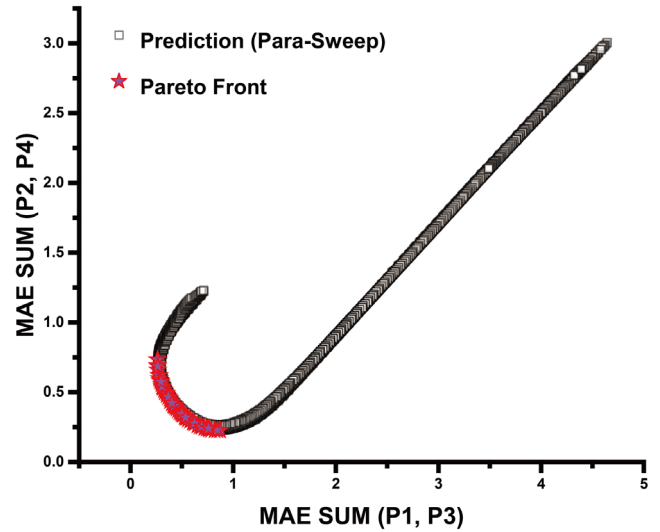


FIGURE 6 | Pareto front visualization showing the trade-off between MAE_SUM (P1, P3) and MAE_SUM (P2, P4) across parameter candidates.

Additionally, Table 2 indicates a 2600x reduction in total simulation time compared with brute-force FEM, highlighting the surrogate’s efficiency.

The surrogate enabled a full search of the 68 452 combinations via NSGA-II, which selected nondominated parameter sets from a hook-shaped Pareto front shown in Figure 6. NSGA-II’s crossover, mutation, and crowding-distance operators ensured diversity while minimizing MAE across all four power conditions [27, 28].

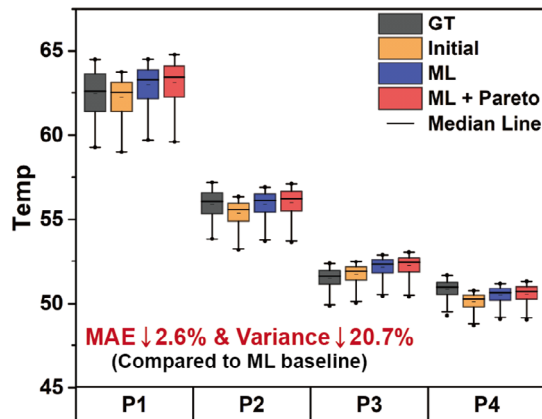
Table 3 shows the convergence of parameters across the three calibration stages—baseline, ML-only, and Pareto-optimized—and that the final values are both physically consistent and well balanced. Compared with ML-only calibration, Pareto optimization reduced average MAE by 2.6% and variance by 20.7%, as shown in Figure 7. Incorporating these parameters into the pipeline enabled the generation of 25 additional FEM datasets, expanding training coverage and improving ML generalization across the full voltage–power range.

2.3 | Heat-Map Prediction via ML Training

Following parameter calibration (Section 3.2), FEM simulations were run for 25 distinct power conditions using the optimized

TABLE 3 | Comparison of thermal parameter values at three stages.

Parameter	Formal sign	Unit	Initial value	ML value	ML + Pareto
Ambient temperature	T_{amb}	K	310	310	310
Heat transfer coefficient	h	W/(m ² ·K)	767	741	738
Thermal conductivity (Si)	k_{Si}	W/(m·K)	212	210	190
Heat capacity (Si)	C_{Si}	J/(kg·K)	684.2	684.2	696.5
Thermal conductivity (Cu)	k_{Cu}	W/(m·K)	400	400	400
Heat capacity (Cu)	C_{Cu}	J/(kg·K)	385	385	385
Surface emissivity	ε	0 to 1	0.7	0.69	0.71

**FIGURE 7** | Box plot comparison of ground truth, initial parameters, ML-optimized parameters, and Pareto-optimized parameters across four power conditions (P1–P4).

settings and combined with four IR-measured profiles to form a 29-sample dataset. Although FEM simulations with Pareto-optimized parameters provide accurate thermal profiles, they rely on linear assumptions and may not fully capture nonlinear factors present in real-world environments. To address this, the four measured power profiles were deliberately incorporated into the dataset, enabling the model to learn from practical thermal behavior, including environmental noise and process-induced uncertainties not accounted for in the simulations.

To ensure that learning is not negatively affected by the small discrepancy ($\sim 0.3^\circ\text{C}$) between the ground truth and the FEM-generated temperatures, the power sweep was carefully designed so that each scenario induces approximately a 1°C difference in the die-level temperature distribution. This deliberate spacing minimizes the impact of residual simulation error, allowing the ML model to focus on learning realistic thermal patterns rather than artifacts from FEM fitting.

Each sample consisted of a surface-temperature map (lid-on) and the corresponding die-level map, both flattened from 61×45 grids (2745 features) and normalized to $[0, 1]$ for stable optimization. The dataset was split into 70% training, 15% validation, and 15% testing.

A lightweight single-layer perceptron (SLP) was selected for its low parameter count, efficient learning on small datasets, and suitability for FPGA deployment. The SLP, as illustrated in

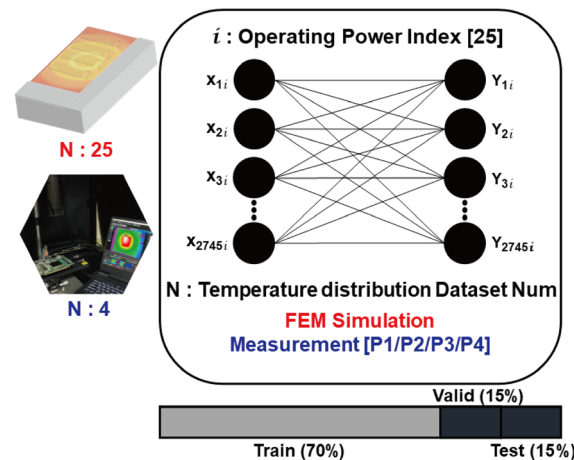
**FIGURE 8** | Overview of the heat-map prediction model using a single-layer perceptron (SLP).

Figure 8, directly mapped surface-temperature vectors (X) to die-temperature vectors (Y). Specifically, both X and Y were defined as 2745-dimensional node vectors expressed in degrees Celsius, where X corresponded to lid-surface temperatures and Y corresponded to internal die temperatures. This direct mapping enabled real-time inference with minimal resource usage.

The performance metrics summarized in Table 4 demonstrate high prediction accuracy on the test set. These results confirmed that combining limited measured data with FEM-generated simulations enables fast and accurate prediction of die-level temperature distribution (heat maps) within advanced packaging systems.

TABLE 4 | Performance comparison of ML-only and Pareto-optimized models on validation and test sets.

Metric	ML-only	ML + Pareto
Validation MAE	0.0122	0.0082
Validation MSE	0.0863	0.0677
Validation RMSE	0.1105	0.0894
Test MAE	0.3591	0.3312
Test MSE	0.1854	0.1565
Test RMSE	0.4480	0.4142

2.4 | FPGA-Based Accelerator Architecture

FPGA-based accelerators achieve superior power efficiency over general-purpose processors by implementing application-specific hardware optimized for massively parallel workloads such as neural network inference. In the Xilinx Zynq series, the programmable logic (PL) region can be configured as a dedicated AI accelerator containing only the essential compute modules and data paths. This selective instantiation minimizes logic overhead, maximizes throughput, and significantly reduces power consumption—attributes that are particularly advantageous for power- and space-constrained platforms such as edge devices and battery-operated systems.

The Zynq architecture integrates an ARM Cortex-A9 processing system (PS) alongside the PL fabric, enabling direct hardware–software performance comparisons within a unified platform. This dual-architecture setup facilitates a codesign approach in which computational workloads can be dynamically partitioned between software execution and hardware acceleration for optimal efficiency.

Figure 9 illustrates the custom hardware architecture developed to execute the trained FC neural network for heat-map prediction. The implementation targeted a Zynq Zybo Z7 board, with model parameters stored on a secure digital (SD) card and loaded into the PS before transfer to the PL via an AXI4 bus. Within the PL, block RAMs (BRAMs) buffered both input vectors and weight parameters. Each parameter was represented in 16-bit fixed-point format, and each 32-bit BRAM word stored two parameters. Four BRAM blocks supported eight parallel FC units, allowing concurrent computation of multiple output nodes.

The FC layer is implemented on the FPGA as a streaming accelerator based on an eight-way fixed-point multiply–accumulate (MAC) array controlled by a lightweight data-mover finite-state machine (FSM). The input feature vector is stored in dual-port BRAM, where each 32-bit word packs two 16-bit fixed-point elements. Weight parameters for eight neurons are distributed

across four BRAM banks using the same packed format, enabling independent and parallel weight access for each MAC pipeline.

During inference, a start command and the input vector length are issued through control registers. The data-mover FSM then sequentially traverses the input BRAM addresses, broadcasting each input word to all MAC units while each MAC unit simultaneously fetches its corresponding weight word from its assigned BRAM bank. Each MAC pipeline performs fixed-point multiplication and accumulation over the full input dimension, with its accumulator retaining the final dot-product result corresponding to one output neuron.

Because all eight MAC units operate concurrently on a shared input stream while consuming distinct weight streams, the architecture produces eight neuron outputs in parallel for each complete pass over the input vector. For networks with a larger output dimension, such as the 2745-neuron heat-map predictor used in this work, outputs are generated iteratively in groups of eight and written back to BRAM for retrieval by the PS.

For benchmarking, the same inference task was executed in software on the ARM Cortex-A9 using identical inputs and parameters. The FPGA-based accelerator substantially outperformed the CPU in both execution time and energy efficiency, demonstrating its suitability for high-throughput, low-power scalar operations such as real-time thermal prediction in advanced packaging systems.

3 | Results and Discussion

3.1 | Model Performance and Heat-Map Visualization

The predictive accuracy and robustness of the proposed heat-map model were evaluated using 29 datasets: 25 FEM-generated thermal profiles—calibrated with optimized parameters—and 4 IR-measured profiles. Performance was assessed under four power

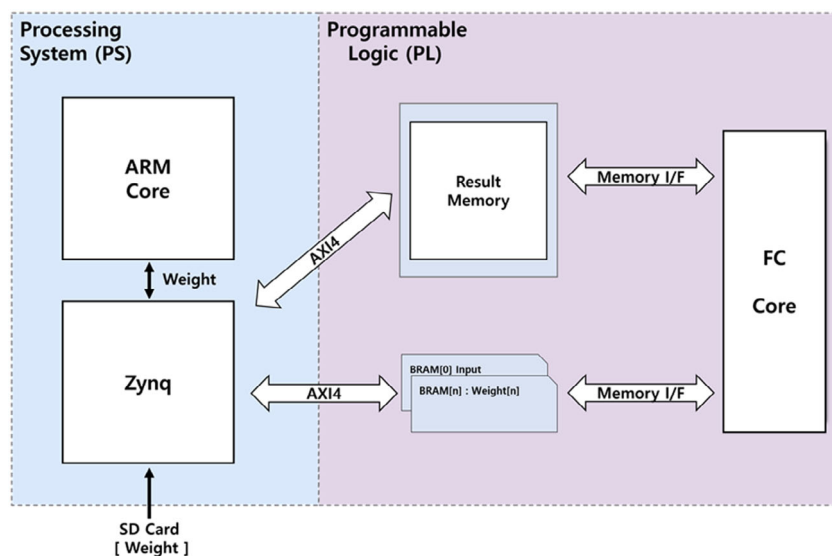


FIGURE 9 | Hardware architecture of the FPGA-based FC neural network accelerator for heat-map prediction. The trained model is stored on an SD card, loaded into the ARM Cortex-A9 PS, and transferred to the PL via an AXI4 bus. BRAMs in the PL buffer input vectors and weight parameters in 16-bit fixed-point format, enabling eight FC units to operate in parallel. Computation results are written back to BRAM for retrieval by the PS, supporting high-throughput, low-power inference for real-time on-chip thermal prediction.

conditions (P1–P4) using both a baseline ML-optimized FEM model and the proposed Pareto-optimized variant.

The Pareto-optimized model consistently outperformed the baseline across all metrics, showing average improvements of 15.9% in MAE, 28.2% in MSE, and 8.1% in RMSE, as summarized in Table 4. The advantage was most pronounced in P4, where the baseline MAE (0.547°C) was reduced to 0.403°C, indicating improved stability in challenging operating conditions. Variance in prediction error across power modes also decreased significantly, demonstrating better generalization.

Figure 10 compares the predicted heat maps against ground-truth measurements across four power conditions (P1–P4), contrasting two model training strategies: (a) the FEM + Pareto

optimization and (b) the baseline ML-only optimization. The Pareto-optimized model accurately captured both localized hotspots and thermal gradients across the die, maintaining uniformly low MAE across all conditions. In contrast, the baseline model performed well under selected scenarios (P1 and P2) but exhibited degraded accuracy in P3 and P4, indicating condition-specific overfitting. This comparison highlights the effectiveness of Pareto-based multiobjective calibration in avoiding performance trade-offs across operating points—a critical feature for real-world thermal management under variable workloads.

Two design choices largely drive these gains. First, Pareto-guided CTM calibration balances errors across P1–P4 instead of

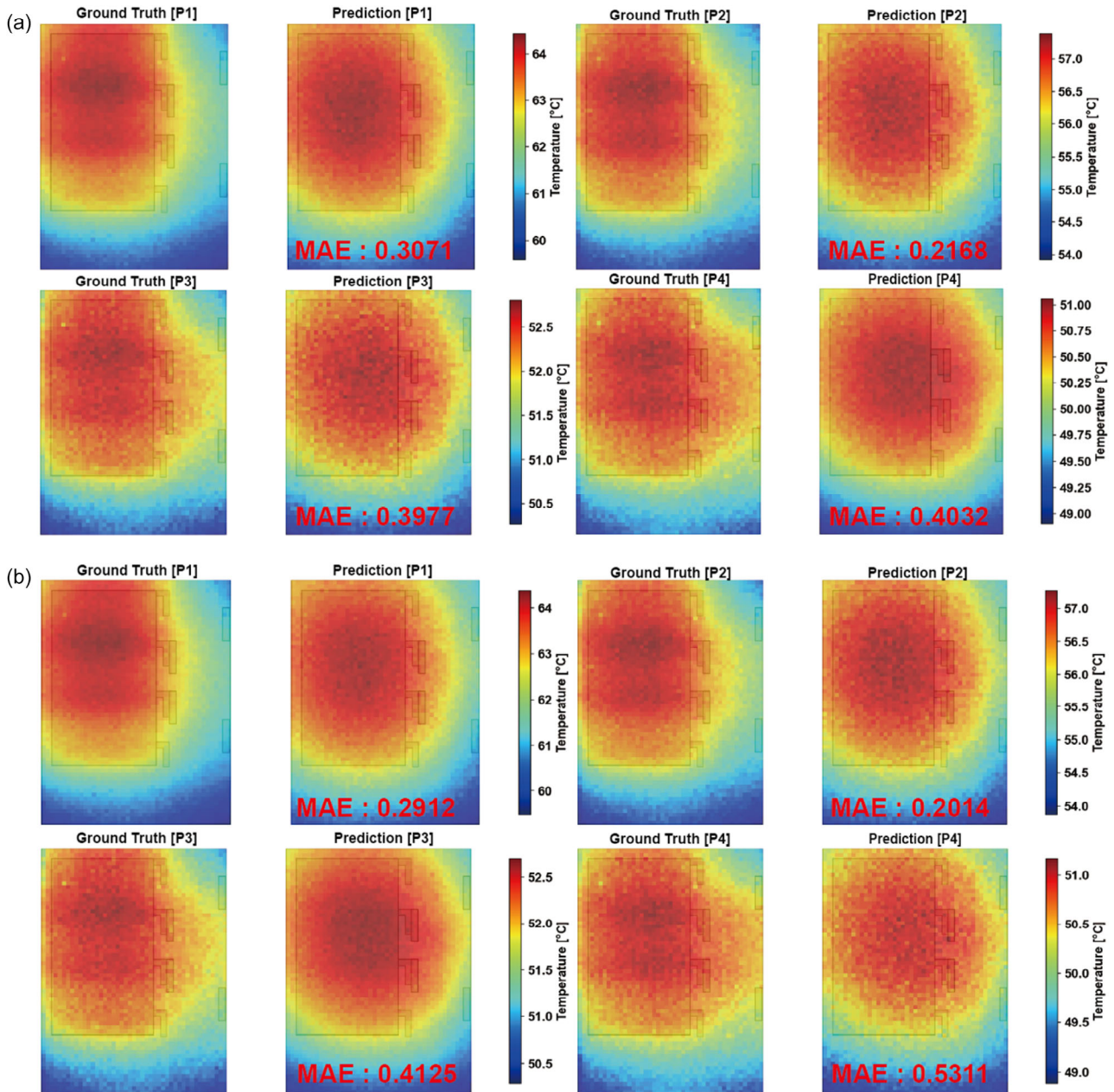


FIGURE 10 | Comparison of predicted die-level heat maps under four power conditions (P1–P4) for two training strategies: (a) FEM + Pareto optimization and (b) baseline ML optimization. Ground truth measurements are shown for reference. The Pareto-optimized model achieves uniformly low prediction errors across all power modes, accurately capturing both localized hotspots and thermal gradients. The baseline model performs well in selected conditions (P1, P2) but shows degraded accuracy in P3 and P4, indicating condition-specific overfitting.

overfitting a single operating point, yielding uniformly low MAE and tighter intercondition variance. Second, the power sweep uses $\sim 1^\circ\text{C}$ spacing between scenarios—larger than the $\sim 0.3^\circ\text{C}$ FEM-measurement residual—so the network learns physically meaningful structure rather than simulation artifacts. Mixing four lid-on/lid-off IR anchors with the synthetic set also injects realistic noise and process variability, boosting robustness. Notably, a lightweight SLP suffices because steady-state conduction imposes a near-linear mapping from lid surface to die field once the CTM is well calibrated; the 61×45 output grid further preserves spatial detail for hotspot localization. In practice, sub- 0.5°C errors at the die level are compatible with proactive controls (e.g., chiplet-level power capping, microbump self-heating), enabling tighter thermal guard bands without sacrificing reliability.

3.2 | FPGA Inference Performance and Energy Efficiency

To validate real-time applicability, the trained SLP model was deployed on a Xilinx Zynq-7000 FPGA (Zybo Z7 board) with the FC layer implemented as a parallel hardware accelerator. The FPGA achieved a core inference time of 0.0143 s—comparable to the 0.0147 s recorded on an RTX 3060 Ti GPU, despite operating at a fraction of the power. In contrast, ARM Cortex-A9 software execution required 8.6408 s, representing a 603-fold speedup. Including data transfer from SD card to BRAM, total FPGA execution time was 1.5005 s (comprising 1.4862 s for data transfer and 0.0143 s for computation), still yielding a 5.76-fold speedup over ARM software execution. The performance and energy-efficiency comparison across FPGA, GPU, and ARM platforms is summarized in Table 5.

Power measurements showed a maximum of 1.71 W, offering $> 20\times$ better energy efficiency than executing the same FC model on a GPU in our benchmarking experiment. The accelerator used eight FC cores operating in parallel, with pipelined execution and four BRAMs for weight and intermediate data storage. A 16-bit fixed-point format minimized resource usage while preserving model accuracy.

These results confirm that the proposed FPGA accelerator delivers GPU-level latency with orders-of-magnitude lower power consumption, making it a strong candidate for deployment in

TABLE 5 | Performance and energy-efficiency comparison of inference execution on FPGA, GPU (RTX 3060 Ti), and ARM Cortex-A9 CPU. The FPGA accelerator achieves GPU-comparable latency at $< 2\text{ W}$ power consumption—over $20\times$ lower than reported GPU/CPU inference workloads—while delivering $> 600\times$ speedup over ARM software execution.

	Data Loading, s	Core, s	End to End, s	Power, W
FPGA HW Core	1.4862	0.0143	1.5005	1.71
ARM Cortex-A9	—	8.6408	8.6408	—
RTX 3060 Ti	—	0.0147	—	36.4

energy- and thermal-constrained intelligent systems such as mobile devices, drones, and edge AI platforms.

From a model-hardware codesign standpoint, the adoption of a lightweight SLP aligns naturally with the structure of the thermal prediction task, which can be formulated as a dense vector-matrix multiplication. This vector-matrix multiplication structure maps naturally onto fixed-point FC cores with BRAM-resident weights, allowing the FPGA implementation to achieve low-latency inference while maintaining ultralow power consumption. Even when including SD-to-BRAM data transfer time, the system sustains an end-to-end latency of 1.5005 s, and practical deployments can further reduce transfer overhead through weight caching or DMA streaming. These results demonstrate that the proposed architecture provides an effective balance of accuracy, latency, and energy efficiency for real-time thermal prediction on resource-constrained edge hardware.

4 | Conclusion

This study demonstrated a power-efficient and scalable framework for real-time die-level temperature prediction in advanced semiconductor packages by integrating calibrated FEM simulation, ML, and FPGA deployment. A compact thermal model was constructed using FEM and refined via hybrid ML-Pareto optimization, enabling high-fidelity predictions from limited empirical data while ensuring balanced accuracy across multiple power conditions.

An SLP-based heat-map predictor, trained on combined simulated and measured datasets, achieved a MAE of 0.3112°C and reduced computational cost by over 2600-fold compared to direct FEM simulation. Implemented as a parallel FC core on a Xilinx Zynq FPGA, the model delivered inference latency comparable to an RTX 3060 Ti GPU while consuming more than 20-fold less power, validating its suitability for energy-constrained environments.

The results confirm that the proposed approach combines high thermal-prediction accuracy with real-time capability and ultralow power operation, making it well suited for mobile computing, edge AI, and IoT systems. Future work will extend the framework to transient thermal prediction and closed-loop thermal management, enabling joint optimization of packaging, power, and performance.

Acknowledgments

This work was supported by Institute for Information & Communications Technology Promotion (IITP) grant funded by the Korea government (MSIT) (RS-2023-00222609, Development of thermal modeling and heat-spreading architecture for heterogeneous integrating package of AI chiplets) and Basic Science Research Program through the National Research Foundation of Korea (NRF) funded by the Ministry of Education (RS-2025-25410048).

Funding

This work was supported by Ministry of Science and ICT, South Korea (RS-2023-00222609) and Ministry of Education RS-2025-25410048).

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

References

1. L. Qian, Y. Xia, X. He, K. Qian, and J. Wang, "Electrical Modeling and Characterization of Silicon-Core Coaxial through-Silicon Vias in 3-D Integration," *IEEE Transactions on Components, Packaging and Manufacturing Technology* 8, no. 8 (2018): 1336–1343, <https://doi.org/10.1109/TCPMT.2018.2854829>.
2. E. Song, K. Koo, J. S. Pak, and J. Kim, "Through-Silicon-Via-Based Decoupling Capacitor Stacked Chip in 3-D-ICs," *IEEE Transactions on Components, Packaging and Manufacturing Technology* 3, no. 9 (2013): 1467–1480, <https://doi.org/10.1109/TCPMT.2013.2257928>.
3. B. W. Ku, K. Chang, and S. K. Lim, "Compact-2D: A Physical Design Methodology to Build Two-Tier Gate-Level 3-D ICs," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems* 39, no. 6 (2020): 1151–1164, <https://doi.org/10.1109/TCAD.2019.2952542>.
4. Y. Yan, H. Yan, S. Yin, L. Zhang, and L. Li, "Single/Multi-Objective Optimizations on Hydraulic and Thermal Management in Micro-Channel Heat Sink with Bionic Y-Shaped Fractal Network by Genetic Algorithm Coupled with Numerical Simulation," *International Journal of Heat and Mass Transfer* 129 (2019): 468–479, <https://doi.org/10.1016/j.ijheatmasstransfer.2018.09.120>.
5. Y. Shin, S. E. Kim, and S. Kim, "Thermal Assessment of Copper through Silicon via in 3D IC," *Microelectronic Engineering* 156 (2016): 2–5, <https://doi.org/10.1016/j.mee.2016.03.018>.
6. D. Song, G. Dong, Y. Yao, Z. Zhu, and Y. Yang, "Optimization and Analysis of Microchannels under Complex Power Distribution in 3-D ICs," *IEEE Transactions on Components, Packaging and Manufacturing Technology* 12, no. 3 (2022): 537–543, <https://doi.org/10.1109/TCPMT.2022.3155148>.
7. L. Zhu, C. Jo, and S. K. Lim, "Power Delivery Solutions and PPA Impacts in Micro-Bump and Hybrid-Bonding 3D ICs," *IEEE Transactions on Components, Packaging and Manufacturing Technology* 12, no. 1 (2022): 45–53, <https://doi.org/10.1109/TCPMT.2021.3125673>.
8. K. Tsai, S. Ku, W. Chang, and H. Tsai, "Thermal Characterization of a Wide I/O 3DIC," in *Proceedings of the 6th International Microsystems, Packaging, Assembly and Circuits Technology Conference (IMPACT)*, (Institute of Electrical and Electronics Engineers (IEEE), 2011): 261–264. <https://doi.org/10.1109/IMPACT.2011.6117256>.
9. L. Cao, C.-C. Wang, C.-Y. Huang, and H.-C. Kuo, "Integrated Design Ecosystem for Chiplets Heterogeneous Integration and Chip-to-Chip Interconnects in Advanced Packaging Technology," in *Proceedings of the IEEE 74th Electronic Components and Technology Conference (ECTC)*, (Institute of Electrical and Electronics Engineers (IEEE), 2024), <https://doi.org/10.1109/ECTC51529.2024.00168>.
10. T. Jiang and H. Hu, "Review of Evolution and Rising Significance of Wafer-Level Electroplating Equipment in Semiconductor Manufacturing," *Electronics* 14, no. 5 (2025): article894, <https://doi.org/10.3390/electronics14050894>.
11. S. Venkateswarlu, S. Mishra, H. Oprins, et al., "Impact of 3-D Integration on Thermal Performance of RISC-V MemPool Multicore SoC," *IEEE Transactions on Very Large Scale Integration (VLSI) Systems* 31, no. 12 (2023): 1896–1904, <https://doi.org/10.1109/TVLSI.2023.3314135>.
12. B. Wunderle, "Thermo-Mechanical Reliability of Flip-Chip Assemblies with Heat-Spreaders," (Ph. D. diss., Technische Universitaet Berlin, 2003), <https://depositonce.tu-berlin.de/bitstreams/0973d70d-fe76-4590-84f0-9bbe461e8e56/download>.
13. H. Oprins, V. Cherman, G. Van der Plas, J. De Vos, and E. Beyne, "Experimental Characterization of the Vertical and Lateral Heat Transfer in Three-Dimensional Stacked Die Packages," *ASME: Journal of Electronic Packaging* 138, no. 1 (2016): 010902-1–010902-9, <https://doi.org/10.1115/1.4032346>.
14. P. K. Chundi, Y. Zhou, M. Kim, E. Kursun, and M. Seok, "Hotspot Monitoring and Temperature Estimation with Miniature On-Chip Temperature Sensors," in *Proceedings of the IEEE/ACM International Symposium on Low Power Electronics and Design (ISLPED)*, (Institute of Electrical and Electronics Engineers (IEEE), 2017): 1–6, <https://doi.org/10.1109/ISLPED.2017.8009191>.
15. W. Huang, K. Skadron, S. Gurumurthi, R. J. Ribando, and M. R. Stan, "Differentiating the Roles of IR Measurement and Simulation for Power and Temperature-Aware Design," in *Proceedings of the IEEE International Symposium on Performance Analysis of Systems and Software*, (Institute of Electrical and Electronics Engineers (IEEE), 2009): 1–10, <https://doi.org/10.1109/ISPASS.2009.4919633>.
16. J. Wen, S. Pan, N. Chang, et al., "DNN-based fast static on-chip thermal solver," in *Proceedings of the 36th Semiconductor Thermal Measurement, Modeling & Management Symposium (SEMI-THERM)*, (Institute of Electrical and Electronics Engineers (IEEE), 2020): 65–75, <https://doi.org/10.23919/SEMI-THERM50369.2020.9142855>.
17. J. Lu, J. Zhang, and S. X.-D. Tan, "Real-time thermal map estimation for AMD multi-core CPUs using transformer," in *Proceedings of the IEEE/ACM International Conference on Computer Aided Design (ICCAD)*, (Institute of Electrical and Electronics Engineers (IEEE), 2023): 1–7, <https://doi.org/10.1109/ICCAD57390.2023.10323817>.
18. M. Żyliński, A. Nassibi, I. Rakhmatulin, A. Malik, C. M. Papavassiliou, and D. P. Mandic, "Deployment of Artificial Intelligence Models on Edge Devices: A Tutorial Brief," *IEEE Transactions on Circuits and Systems II: Express Briefs* 71, no. 3 (2024): 1738–1743, <https://doi.org/10.1109/TCSII.2023.3336831>.
19. V. Peluso, R. G. Rizzo, and A. Calimera, "Efficacy of Topology Scaling for Temperature and Latency Constrained Embedded ConvNets," *Journal of Low Power Electronics and Applications* 10, no. 1 (2020): 10, <https://doi.org/10.3390/jlpea10010010>.
20. Y. Zhang, P. Jiao, W. Du, G. Qi, and B. Chen, "Numerical Simulation of Casting Filling Process Based on SPH-FEM Coupling Method," *Symmetry* 17, no. 4 (2025), article 494, <https://doi.org/10.3390/sym17040494>.
21. M. Oddiraju, Z. Hasnain, and S. Bandyopadhyay, "Physics-Informed Machine Learning towards a Real-Time Spacecraft Thermal Simulator," in *AIAA SciTech Forum*, (2024), <https://doi.org/10.2514/6.2024-4204>.
22. I. Martin-Salinas, J. M. Badia, O. Valls, et al., "Evaluating and Accelerating Vision Transformers on GPU-Based Embedded Edge AI Systems," *The Journal of Supercomputing* 81 (2025): 349, <https://doi.org/10.1007/s11227-024-06807-1>.
23. S. Y. Hosseini Mirmahaleh and M. Y. Hosseini Reshadi, "DLA: Deep learning accelerator," in *Computer Vision and Recognition Systems Using Machine and Deep Learning Approaches: Fundamentals, Technologies and Applications*, ch. 2, (Institution of Engineering and Technology (IET), 2021), 17–49. https://doi.org/10.1049/PBPC042E_ch2.
24. M. L. Wu and J. S. Lan, "Analytical and Finite Element Methodology Modeling of the Thermal Management of 3D IC with through Silicon via," *Soldering & Surface Mount Technology* 28, no. 4 (2016): 177–187, <https://doi.org/10.1108/SSMT-04-2016-0008>.
25. R. B. Bird, W. E. Stewart, and E. N. Lightfoot, *Transport Phenomena**, 2nd (John Wiley & Sons, 2007).
26. COMSOL AB, **COMSOL Multiphysics Heat Transfer Module User's Guide**, ver. 5.5, COMSOL AB (2019), 942–943.

27. M. Arabbeiki, M. Mansourkiaei, D. Ferrero, and A. Castellazzi, "Numerical Optimization of Ejector for Enhanced Hydrogen Recirculation in Proton Exchange Membrane Fuel Cells," *Journal of Power Sources* 598 (2025): Art. no. 233524, <https://doi.org/10.1016/j.jpowsour.2024.233524>.
28. S. M. Shanmugam, M. S. Shunmugam, and K. A. Khan, "Multi Objective Optimization of Friction Stir Welding Parameters Using FEM and Neural Network," *The International Journal, Advanced Manufacturing Technology* 72 (2014): 865–2356, <https://doi.org/10.1007/s00170-014-5727-5>.